



Sneaker Assembly Process Methodology

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Created time	December 9, 2025 4:35 PM
Category	Methodology doc
Last edited by	Laurent
Last updated time	April 22, 2026 12:56 PM
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This document presents the methodology used to construct a sneaker assembly dataset, focused exclusively on the assembly phase of footwear production, using a modular approach by process and by chemical family. It covers assembly process inputs and outputs only and does **not** include physical footwear materials or components, which are modelled separately. The dataset was developed using primary manufacturing data collected through onsite assessment and follow-up data consolidation.

1. Scope and Functional Unit

1.1 System Analysed

The database covers the following assembly processes:

- Lamination
- Cutting
- HF Welding Process
- Screen Printing
- Insole
- Nosew
- Stitching
- Stock fit
- Assembly

Each of these processes was modelled as an autonomous module, with their respective inputs and outputs.

Physical footwear materials and components are outside the scope of this database and are modelled separately.

1.2 Functional Unit

The functional unit adopted is:

- **1 pair of footwear** ("per pair").

The processes were implemented in **Activity Browser** with **reference product** and **amount = 1 unit**, assuming that "unit" represents the **operation equivalent per pair**, consistent with the structure of the model.

2. Modelling Strategy

2.1 Modular logic

The LCI construction followed a layered logic:

1. Primary chemicals per process

(e.g., base adhesives, resins, hardeners, main solvents)

2. Derived/specific chemicals

(hardener, glue, treatment agents, etc.)

3. Cross-cutting inputs and outputs

- energy
- water
- packaging
- NMVOC
- waste outputs

4. Modular integration into the final process activities

(1–9)

This separation enables:

- a clear interpretation of **what enters, where, and why**,
- traceability of **proxy decisions by chemical family**,
- reduced duplication and easier future substitutions

3. Cross-Process Inventory

3.1 Electricity

Electricity supply is modelled in two scenarios:

- **VN:** market for electricity, medium voltage, Vietnam
- **GLO:** market group for electricity, medium voltage, GLO (global average)

The Vietnam scenario was selected as the primary case because it reflects the manufacturing geography represented in the primary data. The GLO scenario provides a location-independent reference. Both are applied consistently across all processes involving electricity consumption.

3.2 Water

Across all processes, water was modelled as:

- **water, decarbonised**

This choice was made to maintain internal consistency and reduce methodological noise until factory- or site-specific industrial water data are available.

3.3 NMVOC emissions

Solvent-related emissions were modelled using:

- **non-methane volatile organic compounds (NMVOC)**
- **compartment: air**
- **Biosphere3**

This approach was kept uniform, assuming emissions associated with solvent-based systems in relevant processes.

3.4 Physical footwear materials and components

No physical materials/components of the footwear were included in this dataset (upper, outsole, foam, knit upper, etc.). The **Assembly Footwear dataset** is a **process-only** dataset covering chemicals, energy, water, packaging, emissions and waste. Material flows are modelled separately.

3.5 Waste outputs

Process-related waste is split into two categories with distinct modelling rules, harmonised across all Carbonfact textile/footwear LCI datasets:

- **Auxiliary / chemical waste** — spent solvents, contaminated rags, cured adhesive and primer residues, off-spec chemicals, mixed municipal solid waste from the process line. These flows are **kept inline inside each process dataset**, as a negative-amount exchange pointing to the appropriate ecoinvent waste-treatment activity (e.g., `treatment of municipal solid waste, open dump, hyperarid infiltration class`, `market for waste polyurethane`). They are **process-specific** and cannot be inferred from a material loss rate.
- **Textile / footwear material waste** — cut-offs of fabric, knit upper, leather, PU sheet, EVA/TPR scraps, upper and outsole trim. These flows are **excluded from the process datasets**. Where such exchanges appear for legacy reasons they are set to **0** with an explanatory comment, for documentation only. Material waste is handled **downstream via the Input-Output Ratio (IOR) chain** (see Section 9), so that one kilogram of gross material input carries its own waste-treatment burden through the material-supplier dataset — avoiding double counting inside the assembly process.

This separation keeps process chemistry/energy intensities comparable across scenarios and electricity grids, and makes material-loss accounting transparent and auditable. In the sneaker database this rule materialises as: the `Outputs - Waste` sub-dataset retains only the mixed-municipal-solid-waste exchange, while the `treatment of waste textile, soiled, municipal incineration` exchange is zeroed — its magnitude is reintroduced at the aggregate level via the IOR.

4. Construction of chemical inputs by phase

4.1 Data source

Chemical quantities per process were calculated using a primary formulation dataset compiled for the reference footwear model.

This dataset contains the list and/or percentage shares of chemicals used by formulation type and by process.

4.2 Calculation method applied

The calculation logic was:

- Identify in Excel the **percentage composition per chemical**, according to:
 - formulation type (hardener, glue, treatment agents, etc.)
 - and associated process (lamination, stock fit, screen printing, etc.)
- Define the **total formulation mass per pair** (or the defined reference mass for each process).
- Convert percentages into absolute quantities by applying:

$$m_i = m_{total} \times \%_i$$

Where:

- m_i = mass of chemical i per pair
- m_{total} = total formulation mass associated with the process
- $\%_i$ = chemical percentage in the Excel dataset
- Add the final values to the inventory for the corresponding process module.

This procedure ensures that:

- quantities reflect the **underlying formulation structure**,
- and the model does not depend solely on generic defaults.

5. Proxies and process-specific decisions

5.1 Lamination

Within lamination chemicals, the modelling approach grouped:

- **Vinyl Acetate – Ethylene copolymers**
- and **Vinyl Acetate**

as represented under the same **vinyl acetate** proxy family for operational consistency within the current dataset version.

5.2 Screen Printing

Proxies were used for resin families, pigments, fillers and solvents, including:

- **polyurethane (flexible foam)** as a resin proxy
- **titanium dioxide + carbon black (50/50)** for pigments
- **brass** for metallic powders
- **silica sand** for texture powders
- **flat glass** for reflective compounds
- **LDPE fine** for matting agent
- **propylene glycol** as humectant
- solvents such as **ethyl acetate** and **n-butyl acetate**

Selection was guided by functional equivalence and compatibility with available ecoinvent entries.

5.3 Stitching

The following proxies were used:

- **SBR rubber** → proxy for synthetic rubber
- **Terpene resin** → proxied by **rosin size**

representing auxiliary materials typically associated with stitching/assembly chemistry.

5.4 Stock fit

Stock fit includes the most complex chemical set and was modelled using carefully selected proxies, including:

- **1-Ethylpyrrolidin-2-one (NEP)** → proxy **NMP**
- **UV-curable resin (urethane acrylate system)** modelled as a representative mixture:
 - urethane acrylate oligomer → proxy PU
 - reactive acrylate diluents → proxy butyl acrylate
 - photoinitiator → chemical, organic
 - plus additional electricity for UV curing

Additionally:

- Methoxypropyl acetate 2 → **DPGME**
- Tetramethylene dimethacrylate → **methyl methacrylate**
- 2-Hydroxyethyl methacrylate → **butyl acrylate**

- Phenol, 4-isocyanato-, phosphorothioate (3:1) → **MDI**

5.5 Assembly (final process)

In the final assembly module, solvent and aromatic acid proxies were applied using functional equivalence logic, including:

- 1,3-dimethylimidazolidin-2-one → **DMF**
- m-toluic acid → **benzoic acid**
- N,N-diethylformamide → **DMF**
- Dimethyl glutarate → **dimethyl malonate**

Cut-off criterion

A formulation component representing **0.10%** of the assembly process input was not included because it falls below the defined relevance threshold.

6. Results (EF v3.1 — Climate Change)

Two electricity scenarios were modelled and compared using **ecoinvent 3.12** (cut-off):

- **VN:** market for electricity, medium voltage, Vietnam
- **GLO:** market group for electricity, medium voltage, GLO (global average)

GWP100 results per process (per pair):

Rank	Process	GWP100 VN (kg CO ₂ -eq)	GWP100 GLO (kg CO ₂ -eq)	Delta
1	8 - Stock fit	0.658	0.425	-35%
2	9 - Assembly	0.423	0.126	-70%
3	7 - Stitching	0.154	0.149	-3%
4	6 - Nosew	0.151	0.060	-60%
5	1 - Lamination	0.133	0.133	~0%
6	2 - Cutting	0.048	0.065	+36%
7	3 - HF Welding	0.058	0.017	-71%
8	4 - Screen Printing	0.039	0.039	~0%
9	5 - Insole	~0	~0	-
	Total	1.664	1.014	-39%

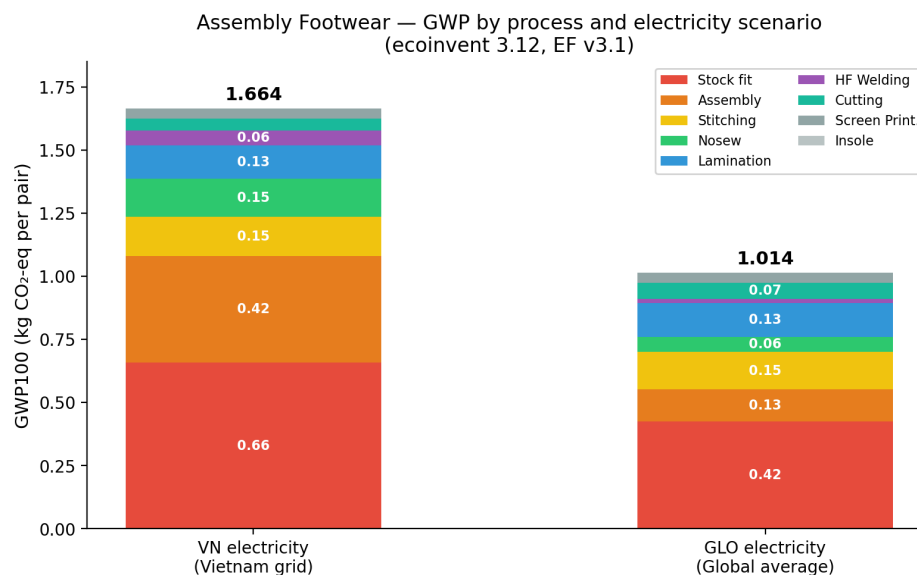


Fig. 1 — GWP100 by process and electricity scenario (ecoinvent 3.12, EF v3.1)

Key findings:

- **Stock fit** is the dominant climate-change driver in both scenarios (~40% of total), driven by chemically intensive bonding/treatment systems and electricity for UV curing.
- Switching from VN to GLO electricity reduces total GWP by **39%** (1.664 → 1.014 kg CO₂-eq).
- Electricity-intensive processes (Assembly –70%, HF Welding –71%, Nosew –60%) are most sensitive to grid choice.
- Chemistry-dominated processes (Lamination, Screen Printing) are virtually unaffected.

6.1 Contribution analysis — Stock fit (VN scenario, 0.658 kg CO₂-eq)

Input	Amount	Unit	GWP (kg CO ₂ -eq)	Share
Electricity (medium voltage, VN)	0.454	kWh	0.233	35.4%
Methyl ethyl ketone	0.028	kg	0.194	29.4%
Ethyl acetate	0.014	kg	0.056	8.5%
Oxalic acid	0.003	kg	0.052	7.9%
Corrugated board box (packaging)	0.027	kg	0.036	5.4%
PU adhesive	0.006	kg	0.030	4.5%
PU flexible foam	0.004	kg	0.024	3.7%
Other (11 inputs)			0.033	5.2%

Stock fit is driven primarily by **electricity** (35%) and **methyl ethyl ketone** (29%). Together these two inputs account for nearly two-thirds of the process footprint. Methyl ethyl ketone (butan-2-one, CAS 78-93-3) has a high emission factor in ecoinvent 3.12 (~7.1 kg CO₂-eq/kg for RoW, ~6.3 for RER) — roughly 2× higher than acetone and 1.8× higher than ethyl acetate — due to petrochemical-intensive production from butane oxidation. The largest single contributor is one treatment-agent formulation (17 g/pair).

6.2 Contribution analysis — Stitching (VN scenario, 0.154 kg CO₂-eq)

Input	Amount	Unit	GWP (kg CO ₂ -eq)	Share
Methyl ethyl ketone	0.009	kg	0.058	37.6%

Input	Amount	Unit	GWP (kg CO ₂ -eq)	Share
Ethyl acetate	0.013	kg	0.053	34.2%
Rosin size	0.003	kg	0.010	6.3%
Synthetic rubber	0.003	kg	0.009	6.0%
Methylcyclohexane	0.003	kg	0.007	4.9%
LDPE granulate (packaging)	0.002	kg	0.007	4.8%
Electricity (medium voltage, VN)	0.010	kWh	0.005	3.4%
Other (3 inputs)			0.004	2.9%

Stitching is dominated by **solvents** (methyl ethyl ketone + ethyl acetate = 72%), with electricity playing a minor role (3.4%). Methyl ethyl ketone alone contributes 38% despite a small mass (9 g/pair), again due to its high emission factor. This solvent-heavy profile explains why switching to GLO electricity barely affects this process (~3%).

7. Limitations and next steps

Item	Status	Next step
Physical footwear materials not included	By design	Material flows modelled separately and coupled later
NM VOC emissions uniform across processes	Simplification	Differentiate by abatement efficiency if data available
Chemical proxies may affect toxicity categories	Known limitation	Refine proxies as more specific datasets become available

8. Summary of methodological decisions

- Modular model by process and chemical families
- FU defined **per pair**
- Two electricity scenarios: VN (market for electricity, medium voltage, Vietnam) and GLO (market group for electricity, medium voltage, GLO)
- Water standardised as **water, decarbonised**
- NM VOC emissions to air via **Biosphere3**
- Proxies applied using functional equivalence logic
- Cut-off applied to **SW-08AT (<1%)**
- **Waste outputs:** auxiliary/chemical waste kept inline (negative exchanges to ecoinvent waste-treatment activities); textile/footwear material waste excluded from process datasets and handled via the IOR chain (Section 9)
- Background database: **ecoinvent 3.12** (cut-off)

9. Treatment of Material Losses, Input–Output Ratios (IOR) and Defect Rate

9.1 Rationale

In parallel to the process-level assembly database, material efficiency losses occurring along the footwear manufacturing chain were quantified using **Input–Output Ratios (IORs)** derived from primary gross and net material consumption data.

This approach allows us to:

- explicitly capture **cutting losses, trimming losses, yield losses and inefficiencies**,
- separate **process chemistry and energy modelling** from **material flow efficiency**,
- and apply losses in a **transparent, auditable and harmonised manner** across the full product system.

9.2 Definition of Input-Output Ratio (IOR)

For each material flow, the Input-Output Ratio was calculated as:

$$\text{IOR} = \frac{\text{Gross material input}}{\text{Net material incorporated into product}}$$

Example:

- Gross material = 105 g
- Net material = 100 g
- IOR = 1.05

An IOR equal to 1 indicates no loss, while values above 1 reflect material losses upstream of the final product.

9.3 Data source and calculation

IORs were computed using the **Materials / Raw Materials table**, based on primary client data, including:

- material specifications,
- gross input quantities per process,
- net quantities effectively incorporated into footwear components.

For each process (lamination, cutting, HF welding, stitching, stock fit, assembly, etc.), individual IORs were calculated at material level.

9.4 Modelling approach and application point

To preserve model clarity and avoid double counting, the following modelling strategy was adopted:

- **All process modules** (chemicals, energy, water, NMVOC, waste) are modelled **per functional unit (1 pair)** assuming ideal yield.
- **Material losses (IORs)** are **not applied within individual process modules**.
- Instead, an **average material IOR**, aggregated across the full process chain, is applied **once at the end of the assembly chain**, on top of the net material demand.

This ensures that:

- chemical and energy intensities remain comparable across scenarios,
- material inefficiencies are accounted for consistently,
- and future updates to loss rates do not require re-building process inventories.

9.5 Defect rate treatment

In addition to process-related material losses, a **defect rate** (rejected or non-conforming pairs) is applied **on top of the average IOR**, reflecting manufacturing quality losses.

The total material correction factor is therefore:

$$\text{Total factor} = \text{Average IOR} \times (1 + \text{Defect rate})$$

This defect rate is applied at the **final product level**, consistent with industrial practice and avoiding distortion of upstream process intensities.

Defect rate: 0.2%

9.6 Assumptions and limitations

- IORs are derived from available primary data and may evolve as additional materials or production lines are integrated.
- Averages are used to balance representativeness and model stability.

- Losses are applied to **physical material flows only**, not to chemicals, energy or water used in assembly processes.
- The approach is conservative and aligned with LCA best practice for complex manufactured products

10. Data Quality Rating (DQR)

10.1 Methodology

DQR follows the ILCD/ecoinvent pedigree matrix with 5 indicators scored 1 (best) to 5 (worst):

- **R (Reliability)**: Quality of data source and verification method
- **C (Completeness)**: Representativeness across relevant sites/factories
- **TeC (Temporal correlation)**: Age of data relative to current year (2026)
- **GC (Geographical correlation)**: Match between data geography and process geography
- **TiC (Technological correlation)**: Match between data technology and modelled technology

DQR = mean(R, C, TeC, GC, TiC)

Data characteristics: The footwear LCI was built from primary bill-of-material data (gross vs. net material quantities) and machine documentation (energy modelled, not metered). The dataset represents one reference footwear model manufactured in Vietnam, with primary data from ~2023.

10.2 DQR Assessment per Sub-process

Dataset	R	C	TeC	GC	TiC	DQR	Notes
Process 1 — Lamination	2	3	1	2	1	1.8	IOR = 1.0. Energy modelled from machine spec
Process 2 — Cutting	2	3	1	2	1	1.8	Multiple materials; IOF from BOM gross/net
Process 3 — HF Welding	2	3	1	2	1	1.8	High IOR (2.154) verified by BOM
Process 4 — Screen Printing	2	3	1	2	1	1.8	Ink not inventoried separately
Process 5 — Insole	3	3	1	2	2	2.2	Data gap: no material loss data
Process 6 — Nosew	2	3	1	2	1	1.8	Small mass contribution
Process 7 — Stitching	2	3	1	2	1	1.8	Lace + heel counter from BOM
Process 8 — Stock Fit	2	3	1	2	1	1.8	Moulded components with trim loss
Process 9 — Assembly	2	3	1	2	1	1.8	Sockliner + adhesive input
Complete Shoe (total)	2	3	1	2	1	1.8	Single reference

Dataset	R	C	TeC	GC	TiC	DQR	Notes
							product configuration

10.3 Key Observations

- **R = 2:** Primary manufacturer BOM data, but energy is modelled from machine documentation rather than metered.
- **C = 3:** Representativeness is limited to a single reference product configuration. Other footwear categories may differ.
- **TeC = 1:** Data from 2023, less than 3 years old.
- **GC = 2:** Vietnam-specific manufacturing data (country-level match).
- **TiC = 1:** Identical technology — data from the exact production process being modelled.
- **Exception:** Insole (Process 5) scores DQR = 2.2 due to data gap (no material quantities available, R=3, TiC=2).

10.4 IOR Summary

The per-process IOR values documented in Section 9 are:

Sub-process	IOR (kg/kg)	Notes
Lamination	1.000	No trim waste
Cutting	1.027	Central value (TPU dominant)
HF Welding	2.154	Highest waste: reflective TPU decorative pieces
Screen Printing	1.027	Substrate from cutting
Insole	1.000	Assumed lossless (data gap)
Nosew	1.142	Label application waste
Stitching	1.122	Lace primary material
Stock Fit	1.163	Outsole dominant (mould trim)
Assembly	1.105	Sockliner trim
Total (complete shoe)	1.088	Gross 0.6915 kg to net 0.6355 kg

11. Reference flow and scope

Reference flow: Assembly Footwear - Process | Functional unit: **1 pair**

This value represents **assembly operations only**, including chemicals, electricity, water, packaging, NMVOC emissions, and process-related waste – **excluding** physical footwear materials and components, which are modelled in a separate system block.

Material inputs, material losses (IOR), and defect rates should be incorporated separately where relevant to the intended assessment boundary, as documented in Section 9.